

FRUHOLMEN FYR, Norway

WMO: 010550

Lat: 71.093N

Lon: 23.982E

Elev: 14

StdP: 101.15

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.2	-7.7	-12.2	1.3	-7.8	-10.9	1.5	-6.4	24.2	2.7	22.1	1.0	10.5	150	0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	17.8	13.4	15.7	12.4	14.0	11.4	14.0	17.1	12.7	15.2	11.7	13.7	5.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.0	15.0	11.2	8.3	13.8	10.3	7.8	12.6	39.3	17.2	36.1	15.4	33.4	13.8	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.0	16.0	DB	-11.0	21.3	2.3	2.7	-12.7	23.3	-14.0	24.8	-15.3	26.3	-17.0	28.3	(4)
(5)			WB	-11.2	16.0	2.4	2.0	-12.9	17.4	-14.3	18.6	-15.6	19.7	-17.3	21.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-1.2	-1.8	-0.8	1.1	4.1	7.0	10.0	10.1	8.3	4.5	1.6	0.2	(6)	
	DBStd	5.03	3.24	3.29	2.83	2.68	2.62	2.42	2.79	2.34	2.57	2.70	2.52	3.14	(7)	
	HDD10.0	2412	348	331	334	267	183	98	34	27	63	170	253	304	(8)	
	HDD18.3	5370	607	564	592	517	440	341	260	254	301	428	503	563	(9)	
	CDD10.0	84	0	0	0	0	1	7	33	31	13	1	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.9	8.8	9.1	9.0	7.6	7.6	7.0	6.3	6.2	7.4	8.2	8.3	9.3	(14)	
(15) Precipitation	PrecAvg	874	91	75	73	53	52	56	64	70	75	90	81	93	(15)	
	PrecMax	1086	158	130	137	95	109	98	152	164	130	145	158	139	(16)	
	PrecMin	669	28	23	24	19	20	17	17	32	29	40	13	45	(17)	
	PrecStd	93	36	27	26	19	23	22	31	29	23	25	33	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.0	6.7	9.7	14.4	18.5	22.1	19.3	16.6	12.3	8.3	7.5	(19)	
		MCWB	4.7	4.1	3.9	5.8	9.2	13.2	15.9	13.9	12.5	9.7	6.6	6.1	(20)	
	2%	DB	5.4	5.0	5.5	6.9	11.3	14.6	18.7	16.9	14.6	10.4	7.2	6.5	(21)	
		MCWB	4.1	3.6	3.8	4.8	7.8	11.3	14.5	13.0	11.9	8.5	5.9	5.3	(22)	
	5%	DB	4.4	4.0	4.2	5.5	9.1	11.8	16.1	15.0	13.1	9.1	6.2	5.5	(23)	
		MCWB	3.2	2.8	2.7	4.1	6.6	9.5	12.8	12.2	10.8	7.7	4.9	4.0	(24)	
	10%	DB	3.3	2.8	3.2	4.7	7.5	9.7	13.9	13.5	11.6	8.3	5.1	4.3	(25)	
		MCWB	1.9	1.4	1.7	3.5	5.5	8.1	11.5	11.3	9.6	7.0	3.8	2.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	4.7	5.1	6.5	10.2	13.7	17.1	14.7	13.6	10.1	7.2	6.7	(27)	
		MCDB	6.0	5.4	5.8	8.5	13.2	17.1	20.8	17.5	15.4	11.5	7.9	7.2	(28)	
	2%	WB	4.4	3.7	4.0	5.2	8.2	11.5	14.6	13.5	12.3	8.6	6.2	5.6	(29)	
		MCDB	5.2	4.7	5.3	6.5	11.0	14.4	18.3	16.3	14.2	9.8	7.0	6.3	(30)	
	5%	WB	3.4	2.8	3.0	4.4	6.8	9.7	13.0	12.5	10.9	7.9	5.1	4.1	(31)	
		MCDB	4.4	3.8	4.2	5.4	8.7	11.6	16.0	14.7	12.8	8.9	6.0	5.2	(32)	
	10%	WB	2.2	1.6	1.9	3.6	5.9	8.3	11.7	11.5	9.9	7.1	4.0	3.0	(33)	
		MCDB	3.2	2.7	3.0	4.6	7.1	9.5	13.7	13.3	11.2	8.2	5.0	4.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.4	2.9	2.8	2.9	2.8	3.4	3.1	2.8	2.6	3.0	3.2	(35)	
		MCDBR	4.1	4.4	3.7	3.5	5.4	6.6	7.9	6.4	4.9	3.5	4.0	4.2	(36)	
	5% WB	MCWBR	3.9	4.1	3.3	2.7	3.3	4.1	4.7	3.9	3.0	2.9	3.7	3.9	(37)	
		MCDWR	4.0	4.3	3.6	3.5	5.0	6.1	7.4	5.9	4.6	3.4	3.9	4.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.265	0.322	0.337	0.347	0.339	0.349	0.337	0.315	N/A	N/A	N/A	(40)	
		taud	N/A	2.077	2.205	2.332	2.422	2.504	2.507	2.525	2.510	N/A	N/A	N/A	(41)	
	Ebn at Noon		0	389	648	785	830	852	827	792	694	N/A	0	0	(42)	
		Edh at Noon	0	36	71	87	91	87	84	73	54	N/A	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.19	1.09	2.81	4.25	4.73	4.20	2.88	1.49	0.38	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.03	0.10	0.22	0.41	0.40	0.44	0.25	0.17	0.05	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	N/A	N/A	N/A	N/A	N/A	-119	-151	N/A	N/A
(47) Regional (10 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page